

CONCEPT 34.7

Humans are mammals that have a large brain and bipedal locomotion (pp. 748–754)

- Derived characters of humans include bipedalism and a larger brain and reduced jaw compared with other apes.
- Hominins**—humans and species that are more closely related to humans than to chimpanzees—originated in Africa about 8 million years ago. Early hominins had a small brain but probably walked upright.
- The oldest evidence of tool use is 2.5 million years old.
- Homo ergaster* was the first fully bipedal, large-brained hominin. *Homo erectus* was the first hominin to leave Africa.
- Neanderthals lived in Europe and the Near East from about 350,000 to 28,000 years ago.
- Homo sapiens* originated in Africa about 195,000 years ago and began to spread to other continents about 180,000 years ago.

? Based on fossil evidence, summarize how key hominin features have changed over time.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Vertebrates and tunicates share
 - jaws adapted for feeding.
 - a high degree of cephalization.
 - an endoskeleton that includes a skull.
 - a notochord and a dorsal, hollow nerve cord.
- Living vertebrates can be divided into two major clades. Select the appropriate pair.
 - the chordates and the tetrapods
 - the urochordates and the cephalochordates
 - the cyclostomes and the gnathostomes
 - the marsupials and the eutherians
- Unlike eutherians, *both* monotremes and marsupials
 - lack nipples.
 - have some embryonic development outside the uterus.
 - lay eggs.
 - are found in Australia and Africa.
- In Figure 34.25, which of the following is the sister taxon to the archosaurs?

(A) synapsids	(C) turtles
(B) crocodilians	(D) lepidosaurs
- As hominins diverged from other primates, which of the following appeared first?

(A) reduced jawbones	(C) the making of stone tools
(B) an enlarged brain	(D) bipedal locomotion

Levels 3-4: Applying/Analyzing

- Which of the following could be considered the most recent common ancestor of living tetrapods?
 - a sturdy-finned, shallow-water lobe-fin whose appendages had skeletal supports similar to those of terrestrial vertebrates
 - an armored, jawed placoderm with two pairs of appendages
 - an early ray-finned fish that developed bony skeletal supports in its paired fins
 - a salamander that had legs supported by a bony skeleton but moved with the side-to-side bending typical of fishes

- EVOLUTION CONNECTION** Living members of a vertebrate lineage can be very different from early members of the lineage, and evolutionary reversals (character losses) are common. Give examples that illustrate these observations, and explain their evolutionary causes.

Levels 5-6: Evaluating/Creating

- SCIENTIFIC INQUIRY • DRAW IT** As a consequence of size alone, larger organisms tend to have larger brains than smaller organisms. However, some organisms have brains that are considerably larger than expected for their size. There are high energetic costs associated with the development and maintenance of brains that are large relative to body size.
 - The fossil record documents trends in which brains that are large relative to body size evolved in certain lineages, including hominins. In such lineages, what can you infer about the costs and benefits of large brains?
 - Hypothesize how natural selection might favor the evolution of large brains despite their high maintenance costs.
 - Data for 14 bird species are listed below. Graph the data, placing deviation from expected brain size on the x-axis and mortality rate on the y-axis. What can you conclude about the relationship between brain size and mortality?

Deviation from Expected Brain Size*	-2.4	-2.1	-2.0	-1.8	-1.0	0.0	0.3	0.7	1.2	1.3	2.0	2.3	3.0	3.2
Mortality Rate	0.9	0.7	0.5	0.9	0.4	0.7	0.8	0.4	0.8	0.3	0.6	0.6	0.3	0.6

Data from D. Sol et al., Big-brained birds survive better in nature, *Proceedings of the Royal Society B* 274:763–769 (2007).

*Values < 0 indicate brain sizes smaller than expected; values > 0 indicate sizes larger than expected.

- WRITE ABOUT A THEME: ORGANIZATION** Early tetrapods had a sprawling gait (like that of a lizard): As the right front foot moved forward, the body twisted to the left and the left rib cage and lung were compressed; the reverse occurred with the next step. Normal breathing, in which both lungs expand equally with each breath, was hindered during walking and prevented during running. In a short essay (100–150 words), explain how the origin of organisms such as dinosaurs, whose gait allowed them to move without compressing their lungs, could have led to emergent properties.

10. SYNTHESIZE YOUR KNOWLEDGE



This animal is a vertebrate with hair. What can you infer about its phylogeny? Identify as many key derived characters as you can that distinguish this animal from invertebrate chordates.

For selected answers, see Appendix A.